

MANAS SHARMA

B.Tech Data Science Undergraduate — Machine Learning & AI Developer

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GitHub Profile: 20+ AI/ML Projects — Research Implementations

SUMMARY

Data Science undergraduate at SRM Institute of Science and Technology specializing in Machine Learning, Reinforcement Learning, Explainable AI, and Intelligent Systems. Experienced in building scalable AI-driven solutions across telecommunications, finance, sustainability, and smart infrastructure domains. IIT Delhi hackathon winner and contributing author of international conference research focused on deep learning optimization and AI-native wireless network automation.

RESEARCH INTERESTS

AI-Native Networks, Reinforcement Learning, Explainable AI, Deep Learning Optimization, Intelligent Communication Systems

TECHNICAL SKILLS

Programming: Python, SQL

Familiar With: Java, JavaScript

Machine Learning: Scikit-learn, XGBoost, Predictive Modeling, Ensemble Learning

AI & Deep Learning: Neural Networks, Computer Vision, NLP, Generative AI, Explainable AI (XAI), Reinforcement Learning

Data Science: Pandas, NumPy, Matplotlib, Statistical Analysis

Tools: Git, Docker, MongoDB, FastAPI, Linux

Concepts: Model Deployment, REST APIs, Optimization Algorithms

EXPERIENCE

Machine Learning Intern 1Stop Chennai Jun 2024 – Jul 2024

- Implemented Random Forest and XGBoost models on datasets containing 10k+ records.
- Improved prediction performance through feature engineering and hyperparameter tuning.
- Evaluated models using accuracy, precision, recall, and cross-validation techniques.
- Contributed to experimentation pipeline and ML workflow validation.

SELECTED PROJECTS

Intent-Driven ATSC Network Slicing (IIT Winner — WOCC 2026 Submission)

- Designed AI-native telecom slicing framework using intent-driven automation.
- Applied reinforcement learning-based optimization for adaptive bandwidth allocation.
- Proposed intelligent resource prioritization for rural broadcasting networks.

6G Intelligent Network Management System

- Developed AI-driven telecom architecture enabling adaptive resource allocation.
- Integrated analytics-based decision support for next-generation network orchestration.

XAI Loan Prediction System

- Built explainable ML model using SHAP for feature attribution and decision transparency.

AI Traffic Management System

- Implemented computer vision pipeline using OpenCV for congestion detection.

Additional AI Projects: Disaster Prediction Analytics, E-Waste Classification AI, A-Q Legal NLP System, GenAI Virtual Try-On Platform

PUBLICATIONS

Reevaluating CNN Filter Dimensions: Experimental Findings on CIFAR-10 and Fashion-MNIST

ICICV 2026 (Scheduled Conference) — Accepted for Oral Presentation

Intent-Driven AI-Native Network Slicing for Rural Broadcasting over ATSC 3.0/B2X

WOCC 2026 — Submission Under Review (EDAS)

EDUCATION

B.Tech in Data Science

SRM Institute of Science & Technology

CGPA: 8.33/10

2022 – Present

ACHIEVEMENTS

- 1st Prize — Build-a-thon 3.0, ITU IIT Delhi
- 1st Prize — Build-a-thon 4.0 ITU
- 2nd Prize — AI Agentic Battle

LANGUAGES

Hindi (Native), English (Professional), German (Beginner)